

Columbia Econ Math Camp 2025

Linear Algebra

Bruno Escobar

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1. Matrix Algebra*

Matrices and their Operations*

Definition 1 (Matrix)

A **matrix** of dimension $m \times n$ is a rectangular array of numbers arranged in m columns and n rows:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Matrices are usually denoted by reference to their **entries** and their dimensions: $\mathbf{A} = [a_{ij}]_{m \times n}$, $\mathbf{A}_{m \times n}$ or $\mathbf{A} \in \mathbb{M}_{m \times n}$ are all common.

- When $n = 1$, $\mathbf{A}$ is called a **column matrix** in $\mathbb{R}^m$; when $m = 1$, it is called a **row matrix** in $\mathbb{R}^n$; if $m = n$, $\mathbf{A}$ is called a **square matrix**, and entries $(a_{11}, a_{22}, \dots, a_{nn})$ are the **diagonal** of $\mathbf{A}$.

Matrices and their Operations*

Let $\mathbf{A} = [a_{ij}]_{m \times n}$, $\mathbf{B} = [b_{ij}]_{m \times n}$, and $\lambda \in \mathbb{R}$:

- The **sum** of two matrices of the same dimensions is another matrix of the same dimensions whose entries are the sums of the corresponding entries of the addend matrices: $\mathbf{C} = \mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]_{m \times n}$ or

$$\mathbf{C} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{12} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

- The **scalar product** of a scalar with a matrix is another matrix where the entries of the original have been multiplied by the scalar:
 $\mathbf{C} = \lambda \mathbf{A} = [\lambda \cdot a_{ij}]_{m \times n}.$

Matrices and their Operations*

Proposition 1

Let $\mathbf{0}_{m \times n} = [0]_{m \times n}$ be the **zero matrix**, and let $\mathbf{A}_{m \times n}$, $\mathbf{B}_{m \times n}$, $\alpha \in \mathbb{R}$, $\beta \in \mathbb{R}$. Some basic properties of addition and scalar multiplication are:

- ❶ $\mathbf{0} + \mathbf{A} = \mathbf{A}$.
- ❷ $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.
- ❸ $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$.
- ❹ $\alpha(\mathbf{A} + \mathbf{B}) = \alpha\mathbf{A} + \alpha\mathbf{B}$.
- ❺ $(\alpha + \beta)\mathbf{A} = \alpha\mathbf{A} + \beta\mathbf{A}$.
- ❻ $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$.
- ❼ $\alpha\mathbf{A} = \mathbf{A}\alpha$.

Matrices and their Operations*

Proposition 2

Let $\mathbf{A}$ and $\mathbf{B}$ be two matrices, $\mathbf{0}$ a zero matrix, and assume conformability where necessary. The following properties of matrix multiplication hold. (Pay attention to 1., 2., and 3.)

- ① Matrix multiplication is **not commutative** in general: $\mathbf{AB}$ is not necessarily equal to $\mathbf{BA}$, even when both products are conformable.
- ② $\mathbf{AB} = \mathbf{0}$ does not necessarily imply that $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$.
- ③ $\mathbf{AB} = \mathbf{AC}$ does not necessarily imply that $\mathbf{B} = \mathbf{C}$.
- ④ It does hold that $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$.
- ⑤ It does hold that $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.
- ⑥ It does hold that $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$.
- ⑦ It does hold that $(\mathbf{A} + \mathbf{B})(\mathbf{C} + \mathbf{D}) = \mathbf{AC} + \mathbf{AD} + \mathbf{BC} + \mathbf{BD}$.

Matrices and their Operations*

Let $\mathbf{A} = [a_{ij}]_{m \times n}$. Four key concepts:

- The **identity matrix** of dimension n , denoted by $\mathbf{I}_n$, is a square matrix whose diagonal entries are all ones and all off-diagonal entries are 0. Its fundamental property is: $\mathbf{A} \cdot \mathbf{I}_n = \mathbf{I}_m \cdot \mathbf{A} = \mathbf{A}$. It is the “1” of matrices.
- If $\mathbf{A}$ is square, the **powers** of $\mathbf{A}$ are $\mathbf{A}^2 = \mathbf{A}\mathbf{A}$, $\mathbf{A}^3 = \mathbf{A}\mathbf{A}\mathbf{A}$, and in general, $\mathbf{A}^k = \underbrace{\mathbf{A}\mathbf{A} \dots \mathbf{A}}_{k \text{ times}}$.
- The **transpose** of $\mathbf{A}$, denoted by $\mathbf{A}'$ or $\mathbf{A}^T$, is obtained by switching rows and columns such that the i -th row of $\mathbf{A}$ becomes the i -th column of $\mathbf{A}'$, and the j -th column of $\mathbf{A}$ becomes the j -th row of $\mathbf{A}'$. Formally, $\mathbf{A}' = [a_{ji}]_{n \times m}$.
- If $\mathbf{A}$ is square, the **trace** of $\mathbf{A}$ is the sum of its diagonal entries:
$$\text{tr}(\mathbf{A}) = a_{11} + a_{22} + \dots + a_{nn} = \sum_{i=1}^n a_{ii}.$$
 Note that the trace is **cyclically commutative**: $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$.

Determinant*

The **determinant** of a square matrix $\mathbf{A}_n$, denoted by $|\mathbf{A}|$ or $\det(\mathbf{A})$, is a well-defined and unique scalar associated with the matrix. The determinant gives us valuable information about the matrix, but mainly we are interested in whether $|\mathbf{A}| = 0$ or not.

Definition 2 (Minors and cofactors)

- The (i, j) **minor matrix** of $\mathbf{A}$ is the submatrix obtained by deleting the i -th row and the j -th column of $\mathbf{A}$.
- The **minor** (i, j) of $\mathbf{A}$, denoted $M_{i,j}$, is the determinant of the (i, j) minor matrix of $\mathbf{A}$.
- The **cofactor** (i, j) of $\mathbf{A}$ is simply $C_{ij} = (-1)^{i+j} M_{i,j}$.

Determinant*

Definition 3 (Determinant)

If $\mathbf{A}$ is a square matrix of order n , the determinant generalizes as:
 $\det(\mathbf{A}) = \sum_{j=1}^n a_{ij} C_{ij}$ for any row i , or equivalently, $\det(\mathbf{A}) = \sum_{i=1}^n a_{ij} C_{ij}$
for any column j .

- If $\mathbf{A}$ is a square matrix of order 2, then $\det(\mathbf{A}) = a_{11}a_{22} - a_{12}a_{21}$.
- If $\mathbf{A}$ is a square matrix of order 3, note that the cofactors of the first

column of $\mathbf{A}$ are $C_{11} = +M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$,

$C_{12} = -M_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$, and $C_{13} = +M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$. Then,

we have that $\det(\mathbf{A}) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$.

NOTE: This is an expansion using the first row of $\mathbf{A}$, but any row or column may be used.

Proposition 3

Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices. The following properties of the determinant are useful (Assume conformability when necessary):

- ① $|\mathbf{A} + \mathbf{B}| \neq |\mathbf{A}| + |\mathbf{B}|$ (i.e., the determinant is not a linear operator).
- ② $|\lambda \mathbf{A}| = \lambda^n |\mathbf{A}|$, where $\lambda \in \mathbb{R}$.
- ③ $|\mathbf{A}| = |\mathbf{A}'|$.
- ④ $|\mathbf{AB}| = |\mathbf{A}| \cdot |\mathbf{B}|$.
- ⑤ If $\mathbf{A}$ is **upper triangular** (i.e., $a_{ij} = 0$ for all $i > j$), **lower triangular** (i.e., $a_{ij} = 0$ for all $i < j$), or **diagonal** (i.e., $a_{ij} = 0$ for all $i \neq j$), then $|\mathbf{A}| = \prod_{i=1}^n a_{ii}$.
- ⑥ If the rows (or columns) of $\mathbf{A}$ form a set of linearly independent vectors, then $|\mathbf{A}| \neq 0$. Otherwise, if the rows (or columns) of $\mathbf{A}$ form a set of linearly dependent vectors, then $|\mathbf{A}| = 0$.

Proposition 4

(Cont'd) Let also $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subset \mathbb{R}^n$. Then:

- ① Let $r \in \mathbb{R}$. If $\mathbf{A} = [r\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$, then $|\mathbf{A}| = r|\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n|$. That is, any scalar r can be factored out of any column (or row) of the determinant.
- ② $|\mathbf{A}| = |\mathbf{v}_1 \ \dots \ \mathbf{v}_i \ \dots \ \mathbf{v}_j \ \dots \ \mathbf{v}_n| = (-1)|\mathbf{v}_1 \ \dots \ \mathbf{v}_j \ \dots \ \mathbf{v}_i \ \dots \ \mathbf{v}_n|$. That is, swapping two columns (or rows) of the determinant only changes the sign.
- ③ If $\mathbf{A} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$, any column (or row) may be replaced by a linear combination of the others and the determinant remains unchanged: $|\mathbf{A}| = |\mathbf{v}_1 - \sum_{i=2}^n (b_i \mathbf{v}_i) \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n|$.

We will later on prove some of these using eigenvalues and eigenvector theory.

Elementary Row Operations*

- As you all probably know, the bread-and-butter of matrix algebra is solving system of linear equations.
- Solving a system in an orderly fashion can be done by way of applying row operations to matrices

Definition 4 (Elementary Row Operations)

The three **elementary row operations (EROs)** to any matrix A of size $n \times m$ with rows $R_1, R_2, \dots, R_n$ are:

- 1 Multiply any row of A by a scalar $\lambda \in \mathbb{R}$: $R_i \leftarrow \lambda \cdot R_i$.
 - 2 Add a row to another more rows: $R_i \leftarrow R_i + R_j$
 - 3 Swap the positions of two rows: $R_i \leftrightarrow R_j$.
- You can also (and definitely should aim to!) do multiple EROs at once.
 - Next, let's see more formally what the EROs are useful for:

Definition 5 (Echelon Form and Pivots)

Let $\mathbf{A}_{m \times n}$ be a matrix. An **echelon form** of $\mathbf{A}$ is another matrix $\mathbf{A}^*$ obtained by applying row operations to $\mathbf{A}$, and that satisfies:

- 1 All rows of zeros are at the bottom of the matrix.
- 2 The first non-zero coefficient of each non-zero row (called a **pivot**) is always strictly to the right of the pivot in the row just above.

Additionally, the **reduced row echelon form** of $\mathbf{A}$ is an echelon form of $\mathbf{A}_r$ that also satisfies:

- 1 The pivot in each non-zero row is 1.
- 2 Every column containing a pivot must have zeros in all other positions.

Definition 6 (Rank)

The **rank** of a matrix $\mathbf{A}_{m \times n}$, denoted $\rho(\mathbf{A})$ or $\text{rank}(\mathbf{A})$, is equal to the number of pivots in any echelon form of $\mathbf{A}$.

- When $\rho(\mathbf{A}) = m$, we say it has **full row rank**. When $\rho(\mathbf{A}) = n$, we say it has **full column rank**.
- If $\mathbf{A}$ is square and $\rho(\mathbf{A}) = n$, we say it has **full rank**.
- Later on, we will learn that an equivalent and more powerful definition of $\rho(\mathbf{A})$ is that is equal to the number of linearly independent columns or rows of $\mathbf{A}$.

Proposition 5

Let $\mathbf{A}$ and $\mathbf{B}$ be matrices. Assume conformability when necessary:

- ❶ $\rho(\mathbf{A}) = \rho(\mathbf{A}')$.
- ❷ $\rho(\mathbf{AB}) \leq \min\{\rho(\mathbf{A}), \rho(\mathbf{B})\}$.
- ❸ As a corollary of 1. and 2., $\rho(\mathbf{A}) = \rho(\mathbf{A}'\mathbf{A})$.
- ❹ If $\mathbf{A}$ is square and has full rank, then $\det(\mathbf{A}) \neq 0$. Conversely, if $\mathbf{A}$ is square and does not have full rank, then $\det(\mathbf{A}) = 0$. It always holds, as a corollary of this definition, that $\rho(\mathbf{A}) \leq \min\{m, n\}$.

Systems of Equations*

Definition 7 (Linear system of equations)

Let $\mathbf{A}_{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{b} \in \mathbb{R}^m$. The equation $\mathbf{Ax} = \mathbf{b}$ is a **linear system** of m equations and n variables, where we aim to solve for $\mathbf{x}$.

- To solve the system, we aim to find a simpler **equivalent system** (i.e., with the same solution set).
- We apply **elementary row operations (ERO)** to the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$ until we reach a convenient echelon form $[\mathbf{A}^* \mid \mathbf{b}^*]$, or its reduced form $[\mathbf{A}_r \mid \mathbf{b}_r]$.
- Applying EROs sequentially always guarantees that $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{A}^*\mathbf{x} = \mathbf{b}^*$ have the same solution(s).
- The process of using EROs to transform the system $\mathbf{Ax} = \mathbf{b}$ into its equivalent $\mathbf{A}_r\mathbf{x} = \mathbf{b}_r$ is called **Gauss-Jordan elimination (GJE)**.

Systems of Equations*

Theorem 1 (Rouché-Capelli)

The reduced row echelon form of any matrix $\mathbf{A}$ is unique. Moreover, let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{x} \in \mathbb{R}^n$. Then the system $\mathbf{Ax} = \mathbf{b}$, with augmented matrix $\mathbf{C} = [\mathbf{A} \mid \mathbf{b}]$ and some echelon matrix $\mathbf{C}^* = [\mathbf{A}^* \mid \mathbf{b}^*]$ satisfies:

- If $\rho(\mathbf{A}^*) < \rho(\mathbf{C}^*)$, then the system has no solution.
- If $\rho(\mathbf{A}^*) = \rho(\mathbf{C}^*) < n$, then there are infinitely many solutions.
- If $\rho(\mathbf{A}^*) = \rho(\mathbf{C}^*) = n$, then there is a unique solution.

Rouché–Capelli implies the following: If $\mathbf{b} = \mathbf{0}$ (in which case we say that the system $\mathbf{Ax} = \mathbf{0}$ is **homogeneous**), then $\rho(\mathbf{A}^*) = \rho(\mathbf{C}^*)$ and therefore:

- (i) There is always at least one solution: $\mathbf{x} = \mathbf{0}$.
- (ii) If $\rho(\mathbf{A}^*) < n$, there are infinitely many solutions.
- (iii) If $\rho(\mathbf{A}^*) = n$, the only solution is $\mathbf{x} = \mathbf{0}$.

Definition 8 (Inverse)

The **inverse** of a square matrix of order n , $\mathbf{A}$, is another unique matrix of the same order, denoted $\mathbf{A}^{-1}$, that satisfies the equalities:

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$$

- The inverse does not always exist. When $\mathbf{A}^{-1}$ does not exist, we say that $\mathbf{A}$ is **singular** or **non-invertible**. When $\mathbf{A}^{-1}$ does exist, we say that $\mathbf{A}$ is **non-singular** or **invertible**.
- The **cofactor matrix** of $\mathbf{A}$ is $\mathbf{C} = [C_{ij}]_{n \times n}$, where C_{ij} is the (i, j) cofactor of $\mathbf{A}$. The cofactor matrix always exists.
- The **adjugate (adjoint) matrix** of $\mathbf{A}$ is simply $\text{adj}(\mathbf{A}) = \mathbf{C}'$.
- Using **cofactor** we can find the inverse of invertible $\mathbf{A}$ as:

$$\mathbf{A}^{-1} = \frac{\text{adj}(\mathbf{A})}{|\mathbf{A}|}$$

Inverse*

The case of the inverse of a 2×2 square matrix $\mathbf{A}$ has a well-known formula. Let's look at it:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Then:

$$\mathbf{C} = \begin{bmatrix} a_{22} & -a_{21} \\ -a_{12} & a_{11} \end{bmatrix}$$

Finally:

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Proposition 6

Let $\mathbf{A}$ and $\mathbf{B}$ be matrices, and $\mathbf{c}$ a vector. Properties of the inverse (Assume both are invertible for properties 3 through 7):

- ① If $\mathbf{A}$ is not invertible, then $\rho(\mathbf{A}) < n$. Moreover, there are infinitely many solutions to the system $\mathbf{Ac} = \mathbf{0}$ besides $\mathbf{c} = \mathbf{0}$.
- ② If $\mathbf{A}$ is invertible, then $\rho(\mathbf{A}) = n$. Moreover, the solution to the system $\mathbf{Ac} = \mathbf{0}$ is unique and given by $\mathbf{c} = \mathbf{0}$.
- ③ $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.
- ④ $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$.
- ⑤ $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.
- ⑥ $(\mathbf{A} + \mathbf{B})^{-1} = \mathbf{A}^{-1}(\mathbf{A}^{-1} + \mathbf{B}^{-1})\mathbf{B}^{-1}$.
- ⑦ $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$.

Special Matrices*

Let $\mathbf{A}$ be a square matrix of order n and $\mathbf{0}_n$ the $n \times n$ matrix of all zeros. Here is a list of special cases of matrices:

- 1 A matrix $\mathbf{A}$ is **symmetric** if $\mathbf{A} = \mathbf{A}'$.
- 2 A matrix $\mathbf{A}$ is **skew-symmetric** if $\mathbf{A} = -\mathbf{A}'$.
- 3 A matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$.
- 4 A matrix $\mathbf{A}$ of order n is **involutory** if $\mathbf{A}^2 = \mathbf{I}_n$.
- 5 A matrix $\mathbf{A}$ is **orthogonal** if its columns (or rows) form an orthogonal set. This implies for square matrices that $\mathbf{A}\mathbf{A}' = \mathbf{A}'\mathbf{A} = \mathbf{I}_n$, and thus, $\mathbf{A}^{-1} = \mathbf{A}'$.
- 6 A matrix $\mathbf{A}$ is **nilpotent** (with **index of nilpotency** k) if there exists $k \in \mathbb{N}$ such that $\mathbf{A}^j = \mathbf{0}_n$ for all $j \geq k$.
- 7 A matrix $\mathbf{A}$ of order n is **sequentially stable** if $\lim_{k \rightarrow \infty} \mathbf{A}^k = \mathbf{0}_n$.

Summary*

To summarize the review, for a square matrix $\mathbf{A}$ of order n , and $\mathbf{x} \in \mathbb{R}^n$, the following statements are equivalent:

- 1 $|\mathbf{A}| \neq 0$.
- 2 $\mathbf{A}$ has full rank, $\rho(\mathbf{A}) = n$.
- 3 (After learning about linear independence:) The columns and rows of $\mathbf{A}$ are linearly independent.
- 4 $\mathbf{A}$ is non-singular, its inverse $\mathbf{A}^{-1}$ exists and is unique.
- 5 The linear system $\mathbf{Ax} = \mathbf{0}$ has a unique solution $\mathbf{x} = \mathbf{A}^{-1}\mathbf{0} = \mathbf{0}$.

The following statements are also equivalent:

- 1 $|\mathbf{A}| = 0$.
- 2 $\mathbf{A}$ does not have full rank, $\rho(\mathbf{A}) < n$.
- 3 (After learning about linear independence:) The columns and rows of $\mathbf{A}$ are linearly dependent.
- 4 $\mathbf{A}$ is singular, its inverse $\mathbf{A}^{-1}$ does not exist.
- 5 The linear system $\mathbf{Ax} = \mathbf{0}$ has infinitely many solutions.

2. Vectors and Vector Spaces

Definition 9 (Vector sapce)

A triple $(V, +, \cdot)$ composed of

- set V , whose elements are called **vectors**.
- an operation $+: V^2 \rightarrow V$ called **vector addition**.
- an operation $\cdot: \mathbb{R} \times V \rightarrow V$ called **scalar multiplication**.

is said to be a **(real) vector space** iff it satisfies these 7 axioms:

- 1 $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ and $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- 2 There exists a **zero vector** $\mathbf{0}$ s.t. $\mathbf{0} + \mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{u}$, $\forall \mathbf{u} \in V$.
- 3 Any $\mathbf{u} \in V$ has an **additive inverse**, or $-\mathbf{u}$, s.t. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
- 4 There exists 1, a multiplicative identity in $\mathbb{R}$, s.t. $1 \cdot \mathbf{u} = \mathbf{u}$.
- 5 For any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\mathbf{v} \in V$, we have $(\lambda_1 \lambda_2) \mathbf{v} = \lambda_1 (\lambda_2 \mathbf{v})$.
- 6 $\forall \lambda \in \mathbb{R}$, $\mathbf{v}_1, \mathbf{v}_2 \in V$, we have $\lambda(\mathbf{v}_1 + \mathbf{v}_2) = \lambda \mathbf{v}_1 + \lambda \mathbf{v}_2$.
- 7 $\forall \lambda_1, \lambda_2 \in \mathbb{R}$, $\mathbf{v} \in V$, we have $\mathbb{R}$: $(\lambda_1 + \lambda_2) \mathbf{v} = \lambda_1 \mathbf{v} + \lambda_2 \mathbf{v}$.

Vector Spaces

- In general the operations $+$ and $\cdot$ do not have to be the same as their analogous on the real line.
 - ▶ Although you can trivially show that $\mathbb{R}$ with simple sum and multiplication operations are a vector space.
- The Euclidean vector space $(\mathbb{R}^n, +, \cdot)$ with $\mathbf{u} + \mathbf{v} := (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$ and $\lambda \cdot \mathbf{v} := (\lambda v_1, \lambda v_2, \dots, \lambda v_n)$ is also a vector space.
- We can define a complex vector space; this will be useful when computing complex-valued eigenvectors.
- We can define vector spaces over very general objects. For example, the set of functions from X to $\mathbb{R}$, $\mathcal{F}$, with $(f + g)(x) := f(x) + g(x)$ and $(\lambda \cdot f)(x) := \lambda f(x)$ is a vector spaces.

Exercise 1

Show that the zero vector and additive inverse of any $\mathbf{v} \in V$ are unique. Show $0 \cdot \mathbf{v} = \mathbf{0}$ for any $\mathbf{v} \in V$ and $\lambda \cdot \mathbf{0} = \mathbf{0}$ for any $\lambda \in \mathbb{R}$.

Definition 10 (Vector Subspaces)

Given a vector space $(V, +, \cdot)$, a subset $W \subset V$ is said to be a **vector subspace** of V iff

- 1 W contains the zero vector $\mathbf{0} \in V$.
- 2 W is **closed under addition**: $\forall \mathbf{u}, \mathbf{v} \in W$ we have $\mathbf{u} + \mathbf{v} \in W$.
- 3 W is **closed under scalar multiplication**: $\forall \mathbf{u} \in W, \lambda \in \mathbb{R}$, we have $\lambda \cdot \mathbf{u} \in W$.

- Note that vector spaces satisfy these properties trivially.
- Intuitively, a subset W will only fail to be a vector (sub)space when the result of a basic operation in W does not belong to W .

Exercise 2

Show that any intersection of vector subspaces is a vector subspace.

Definition 11 (Span)

Let $(V, +, \cdot)$ be a vector space, $S \subset V$. The **linear span** of S is the smallest vector subspace that contains S , or more concretely the intersection of all subspaces of V that contain S .

$$\text{Span}(S) = \bigcap \{W : W \text{ is a vector subspace of } V, S \subset W\}$$

We also call it the **vector subspace spanned/generated by S** .

- As an example, the first quadrant of $\mathbb{R}^2$ is not a vector subspace.
- However, we can add some elements to it (in fact, the second to fourth quadrants) to obtain its span, which is $\mathbb{R}^2$.
- The span is always well defined and is guaranteed to be a vector space, since we proved in the last slide that an arbitrary intersection of vector subspaces is also a vector subspace.

Vector spaces

We will often be interested in finding span of finite sets. For that we need to know what a linear combination is.

Definition 12 (Linear Combinations)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \in V$. A **linear combination** of S is a weighted sum of vectors in S , i.e. $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$, where $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$.

For finite set of vectors, it can be shown that $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is simply the set of all linear combinations of these n vectors. Finding this set is simply a GJE exercise

Exercise 3

Show that $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is the set of all linear combinations of the vectors.

Linear Dependence

Sometimes some vectors add no additional information to a spanning set, i.e. when $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \text{span}(\mathbf{v}_1, \mathbf{v}_3)$. This is related to the concept of linear dependence.

Definition 13 (Linear Dependency)

In vector space $(V, +, \cdot)$, a finite set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is said to be **linearly independent (L.I.)** iff $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n) = \mathbf{0}$ is the unique solution to the system of equations $\sum_{i=1}^n \lambda_i \mathbf{v}_i = \mathbf{0}$, also known as a **linear contrast**. Otherwise, that is if there is a non-zero solution to the linear contrast, we say $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are **linearly dependent (L.D.)**.

Linear Dependence

Linear dependence is easier to understand with this equivalent definition.

Proposition 7

In vector space $(V, +, \cdot)$, a finite set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ with $n \geq 2$ are L.D. iff some $\mathbf{v}_i$ can be represented as a linear combination of the other $\{\mathbf{v}_j\}_{j \neq i}$.

- Intuitively, in an L.D. set, some vector contains some redundant information when generating the span.
- L.I. vectors all have unique or non-redundant information. This is reflected in the next proposition.

Proposition 8

Let $\mathbf{u} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ are L.I. Then $\exists! \lambda_1, \lambda_2, \dots, \lambda_n$ s.t. $\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$

Definition 14 (Basis)

Let S be a vector subspace of vector space $(V, +, \cdot)$. The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in S$ are a **(finite) basis** of S iff

- ① $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are L.I. and
- ② $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ span S .

- A basis of S is a set of vectors that can span S and that have no redundant information.
- Basis are not unique! In fact, generally there are infinitely many sets of vectors that span any given set.
- But even though there are infinitely many bases, they all share one property: They must have the same number of vectors!

Proposition 9

In vector space $(V, +, \cdot)$, if vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m$ can each be represented by some linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and $m > n$, then $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m$ are L.D.

With this, you can show that the number of vectors in a basis of S must be unique, and we call it the **dimension** of S , or $\dim(S)$.

Theorem 2 (Dimension)

In vector space $(V, +, \cdot)$, $S \subset V$. These three are equivalent:

- 1 Any basis of S has n elements.
- 2 n is the maximum number of L.I. vectors of S .
- 3 n is the minimum number of vectors in S that can represent all vectors in S .

Basis

As an immediate corollary of the dimension theorem, we get a super useful proposition:

Proposition 10

In vector space $(V, +, \cdot)$, let $S \subset V$ be a vector subspace. Then $\dim(S) = n$ iff any set of n vectors in S that are L.I. is basis of S .

- It is easy to see that any n vectors in $\mathbb{R}^n$ that are L.I. must be a basis of $\mathbb{R}^n$.
- Finally, I won't show this in general but basis exist for every vector space, even infinite-dimensional ones.
- This is hard to prove in some vector spaces (e.g. real valued functions), but it is trivial for subspaces of known finite-dimensional spaces (try it at home!).

Inner Products and Norms

Definition 15 (Inner product spaces)

Let $(V, +, \cdot)$ be a vector space. The 4-tuple $(V, +, \cdot, \langle \cdot, \cdot \rangle)$ is an **inner product space** iff the **inner product operator** $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfies the following properties:

- (1) **Commutativity:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ for any $\mathbf{u}, \mathbf{v} \in V$,
- (2) **Linearity:** $\langle \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2, \mathbf{v} \rangle = \lambda_1 \langle \mathbf{u}_1, \mathbf{v} \rangle + \lambda_2 \langle \mathbf{u}_2, \mathbf{v} \rangle$ for any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\mathbf{u}_1, \mathbf{u}_2, \mathbf{v} \in V$, and
- (3) **Positive definiteness:** $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ for any $\mathbf{u} \in V$, and equality holds iff $\mathbf{u} = \mathbf{0}$.

- We say $\mathbf{v}, \mathbf{u} \in V$ are **orthogonal** iff $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.
- The classic example of inner product operator is the **dot product** in $\mathbb{R}^n$ defined as $\mathbf{x} \cdot \mathbf{y} := \sum_{i=1}^n x_i y_i$. At home show this satisfies all three properties.

Inner Products and Norms

The following result is very useful:

Theorem 3 (Cauchy-Schwarz inequality)

Let $(V, +, \cdot, \langle \cdot, \cdot \rangle)$ be an inner product space, $\|\mathbf{u}\| := \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$. For any $\mathbf{u}, \mathbf{v} \in V$, we have that

$$\|\mathbf{u}\| \|\mathbf{v}\| \geq |\langle \mathbf{u}, \mathbf{v} \rangle|$$

The next result might prove useful in some proofs as well.

Theorem 4 (Minkowski inequality)

Let $p \geq 1$ and $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{y} = (y_1, \dots, y_n) \in \mathbb{R}^n$. Then

$$\left(\sum_{i=1}^n |x_i + y_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} + \left(\sum_{i=1}^n |y_i|^p \right)^{1/p}.$$

Inner Products and Norms

Definition 16 (Normed spaces)

Let $(V, +, \cdot)$ be a vector space. The 4-tuple $(V, +, \cdot, \|\cdot\|)$ is a **normed (vector) space** iff the **norm function** $\|\cdot\| : V \rightarrow \mathbb{R}_+$ satisfies the following properties:

- (1) $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = 0$, for any $\mathbf{v} \in V$,
- (2) $\|\lambda\mathbf{v}\| = |\lambda|\|\mathbf{v}\|$, for any $\lambda \in \mathbb{R}$ and $\mathbf{v} \in V$, and
- (3) **Triangle inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, for any $\mathbf{u}, \mathbf{v} \in V$.

- Crucially, given an inner product space $(V, +, \cdot, \langle \cdot, \cdot \rangle)$, we can always **induce a norm** using the inner product by defining $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.
- The most common norm function is the **Euclidean norm**, which is the result of inducing a norm using the dot product in $\mathbb{R}^n$.
- In turn, we can also use the Euclidean norm to induce the Euclidean metric as $d_2(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$.

Inner Products and Norms

More formally, the set of norms induced by inner products is a subset of the set of all norms.

Proposition 11

Let $(V, +, \cdot, \langle \cdot, \cdot \rangle)$ be an inner product space, and define its induced norm $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$. Then $(V, +, \cdot, \|\cdot\|)$ is a normed vector space.

Similarly, the set of metrics induced by norms (and more so the set of metrics induced by induced norms) is a subset of the set of all metrics.

Proposition 12

Let $(V, +, \cdot, \|\cdot\|)$ be a normed space, and define its induced metric as $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|$. Then (V, d) is a metric space.

3. Eigenvalues, Eigenvectors and Diagonalization

Eigenvalues and eigenvectors

The concept of eigenvalues appears often in economic models and econometrics, and is particularly useful in dynamic analysis

Definition 17 (Eigenvalue and eigenvector)

Let $\mathbf{A}$ be an $n \times n$ matrix over $\mathbb{C}$. We say that $\lambda \in \mathbb{C}$ is an **eigenvalue** if there exists a non-zero complex vector $\mathbf{x} \in \mathbb{C}^n - \{\mathbf{0}\}$ s.t. $\mathbf{Ax} = \lambda\mathbf{x}$ and we call $\mathbf{x}$ an **eigenvector** of $\mathbf{A}$ associated to eigenvalue λ .

- The set of eigenvalues of $\mathbf{A}$ is called the **spectrum**, or $\sigma(\mathbf{A})$.
- Crucially an eigenvector $\mathbf{x} \neq \mathbf{0}$, otherwise any scalar is an eigenvalue.
- The following are equivalent ways of saying $\lambda \in \mathbb{C}$ is an eigenvalue:
 - 1 System of equations $(\mathbf{A} - \lambda I_n)\mathbf{x} = \mathbf{0}$ has non-zero solution.
 - 2 The columns of matrix $\mathbf{A} - \lambda I_n$ are L.D.
 - 3 $P_{\mathbf{A}}(\lambda) := \det(\mathbf{A} - \lambda I_n) = 0$.

Eigenvalues and eigenvectors

- Expanding $\det(\mathbf{A} - \lambda I_n)$ always results in a *polynomial of degree n* of λ . Hence we name $P_{\mathbf{A}}(\lambda)$ the **characteristic polynomial** of $\mathbf{A}$.
- Then the spectrum is simply the set of all roots of $P_{\mathbf{A}}(\lambda)$, which we know have exactly n roots after counting multiplicities due to a famous result.

Theorem 5 (Fundamental Theorem of Algebra)

Let $P : \mathbb{C} \rightarrow \mathbb{C}$ be a **polynomial of degree n** , i.e. $P(\lambda) = c_n \lambda^n + c_{n-1} \lambda^{n-1} + \dots + c_1 \lambda + c_0$ where $\{c_k\}_{k=1}^n \in \mathbb{C}$ and $c_n \neq 0$. Then P has exactly n **roots** in $\mathbb{C}$, *after counting multiplicity*. That is, there exist scalars $\{\lambda_k\}_{k=1}^m \in \mathbb{C}$ with some corresponding naturals $\{a_k\}_{k=1}^m \in \mathbb{N}$, with $m \leq n$ and $\sum_{k=1}^m a_k = n$, such that

$$P(\lambda) = c_n (\lambda - \lambda_1)^{a_1} (\lambda - \lambda_2)^{a_2} \dots (\lambda - \lambda_k)^{a_k}.$$

We say that a_k is the **algebraic multiplicity** of root λ_k .

Eigenvalues and eigenvectors

- Note that, as with any polynomial, a given eigenvalue may be a root *or more than one root*. This will matter when we learn about diagonalization.
- Nevertheless, if the spectrum of $\mathbf{A}$ is $\sigma(\mathbf{A}) = \{\lambda_k\}_{k=1}^m$ with algebraic multiplicities $\{a_k\}_{k=1}^m$, then clearly $P_{\mathbf{A}}(\lambda) = \prod_{k=1}^m (\lambda - \lambda_k)^{a_k}$.
- From this formula a couple useful results emerge already.

Proposition 13

Let $\mathbf{A}$ be an $n \times n$ matrix over $\mathbb{C}$ with spectrum $\sigma(\mathbf{A}) = \{\lambda_k\}_{k=1}^m$ and algebraic multiplicities $\{a_k\}_{k=1}^m$. Then

- 1 $\det(\mathbf{A}) = \prod_{k=1}^m \lambda_k^{a_k}$, and
- 2 $\operatorname{tr}(\mathbf{A}) = \sum_{k=1}^m a_k \lambda_k$.

Quick but important corollary: $|\mathbf{A}| = 0$ iff $0 \in \sigma(\mathbf{A})$.

Diagonalization

Now let's learn a concept of matrices that is useful due to handful of properties that come with it.

Definition 18 (Diagonalization)

Let $\mathbf{A}$ be an $n \times n$ matrix over $\mathbb{C}$. We say that $\mathbf{A}$ is **diagonalizable** in $\mathbb{R}$ (in $\mathbb{C}$) if there exists some invertible matrix $\mathbf{P}$ over $\mathbb{R}$ ($\mathbb{C}$) and an $n \times n$ diagonal matrix Λ over ($\mathbb{R}$) $\mathbb{C}$ s.t. $\mathbf{A} = \mathbf{P}\Lambda\mathbf{P}^{-1}$. The right hand side $\mathbf{P}\Lambda\mathbf{P}^{-1}$ is also called the **spectral decomposition** of $\mathbf{A}$.

- Some textbooks define the concept of **similar** (square) matrices $\mathbf{A}, \mathbf{B}$ when we can represent $\mathbf{A} = \mathbf{Q}\mathbf{B}\mathbf{Q}^{-1}$ for some invertible $\mathbf{Q}$.
- Thus $\mathbf{A}$ is diagonalizable iff it is similar to some *diagonal* matrix Λ . Similarity ensures $\mathbf{A}$ inherits many properties from Λ .
- Not every square matrix is diagonalizable in $\mathbb{R}$ or even in $\mathbb{C}$, we will see many examples.

Diagonalization

A crucial thing that we must realize is that, if a matrix $\mathbf{A}$ is diagonalizable, then its similar diagonal matrix must be populated by its eigenvalues:

$$\begin{aligned}P_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda I_n) &= \det(\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1} - \lambda I_n) \\&= \det(\mathbf{P}(\mathbf{\Lambda} - \lambda I_n)\mathbf{P}^{-1}) \\&= \det(\mathbf{P}) \det(\mathbf{\Lambda} - \lambda I_n) \det(\mathbf{P}^{-1}) \\&= \det(\mathbf{\Lambda} - \lambda I_n) = P_{\mathbf{\Lambda}}(\lambda)\end{aligned}$$

- Since they have the same characteristic polynomial, then *both $\mathbf{A}$ and $\mathbf{\Lambda}$ must have the same eigenvalues* (with same alg multiplicities).
- But since $\mathbf{\Lambda}$ is a diagonal matrix, this clearly implies that its diagonal elements must be equal to the eigenvalues of $\mathbf{A}$.
- Note that $\mathbf{\Lambda}$ is not unique as we can order the eigenvalues in any way.
- A simple but crucial corollary is that matrix $\mathbf{P}$ must be composed of n (L.I.) eigenvectors of $\mathbf{A}$, more specifically, the i -th column of $\mathbf{P}$ is an eigenvector of $\mathbf{A}$ associated to the i -th element of the diagonal of $\mathbf{\Lambda}$.

Diagonalization

The invertibility of $\mathbf{P}$ and the insights from the previous slides hint strongly at the conditions for diagonalization, which we state next

Theorem 6 (Diagonalizability)

Let $\mathbf{A}$ be an $n \times n$ matrix over $\mathbb{C}$. Then $\mathbf{A}$ is diagonalizable over $\mathbb{C}$ iff there exists a set of n eigenvectors of $\mathbf{A}$ that are L.I.

- Some textbooks define this notion as follows.
- Define $\mathcal{E}_{\mathbf{A}}(\lambda) := \{\mathbf{x} \in \mathbb{C}^n : \mathbf{A}\mathbf{x} = \lambda\mathbf{x}\}$, which in words is the (vector subspace) set with all eigenvectors of $\mathbf{A}$ associated to eigenvalue λ (plus the $\mathbf{0}$), we call this the **eigenspace** of $\mathbf{A}$ for eigenvalue λ .
- Define the **geometric multiplicity** of eigenvalue λ_k as $g_k := \dim(\mathcal{E}_{\mathbf{A}}(\lambda_k))$, which, intuitively, is just the size of the largest subset of L.I. vectors from $\mathcal{E}_{\mathbf{A}}(\lambda_k)$.
- Then a matrix is diagonalizable iff the algebraic and multiplicities of each eigenvalue λ_k coincide, i.e. $a_k = g_k$.

Diagonalization

- This equivalence should make sense, because in order to make $\mathbf{P}$ invertible, we need to construct n L.I. vectors matched with each of the n eigenvalues.
- Eigenvectors from different eigenspaces cannot possibly be represented as linear combinations of each other (otherwise they would not be eigenvectors of their respective eigenvalues).
- Then we just need to make sure that for each eigenvalue, we can build as many L.I. eigenvectors as required by its algebraic multiplicity.
- If an eigenvalue λ_k has $a_k = 1$ then it follows immediately that $g_k = 1$, and thus the next proposition follows easily

Proposition 14

Let $\mathbf{A}$ be an $n \times n$ matrix over $\mathbb{C}$. If $\mathbf{A}$ has n distinct eigenvalues (i.e. $\lambda_i \neq \lambda_j$ if $i \neq j$) then $\mathbf{A}$ is diagonalizable over $\mathbb{C}$.

Diagonalization

- Recall that an $n \times n$ matrix $\mathbf{A}$ is **orthonormal** if $\mathbf{A}^T \mathbf{A} = \mathbf{A} \mathbf{A}^T = \mathbf{I}_n$, or equivalently, if $\mathbf{A}^{-1} = \mathbf{A}^T$ which implies that its columns are pairwise orthogonal and each has a norm of 1 (w.r.t dot product).
- With that we can write down the most famous theorem on diagonalization. I will not prove part (i), but I can do part (ii).

Theorem 7 (Spectral theorem)

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix over $\mathbb{R}$. Then

- (i) All its eigenvalues are real and $\mathbf{A}$ is diagonalizable over $\mathbb{R}$.
- (ii) Further there exists an orthonormal matrix $\mathbf{P}$ s.t.
$$\mathbf{A} = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^T.$$

4. Quadratic Forms

Quadratic Forms

Definition 19 (Quadratic Forms)

A **quadratic form** on $\mathbb{R}^n$ is a function $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ that can be represented as $Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ where $\mathbf{A}$ is some real $n \times n$ matrix.

These quadratic forms often appear in optimization second order conditions and statistical variances. You may recall them as:

$$\begin{aligned} Q(\mathbf{x}) &= \mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \\ &= a_{11}x_1^2 + a_{22}x_2^2 + \cdots + a_{nn}x_n^2 \\ &\quad + (a_{12} + a_{21})x_1x_2 + (a_{13} + a_{31})x_1x_3 \\ &\quad + \cdots + (a_{n-1,n} + a_{n,n-1})x_{n-1}x_n \end{aligned}$$

For any quadratic form Q , infinitely many matrices can represent it, but *there is only one symmetric matrix $\mathbf{A}$ that represents it*, so we will talk interchangeably about Q and its representative symmetric matrix $\mathbf{A}$.

Quadratic Forms

We can talk about the **sign** of a matrix as follows, which is often used in second order conditions

Definition 20 (Matrix definiteness)

Let $\mathbf{A}$ be an $n \times n$ real symmetric matrix. The matrix $\mathbf{A}$, and the quadratic form $Q(\mathbf{x}) := \mathbf{x}^T \mathbf{A} \mathbf{x}$ that is represented by $\mathbf{A}$, are called...

- ① ... **positive definite**, iff $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$ for any $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$;
- ② ... **negative definite**, iff $\mathbf{x}^T \mathbf{A} \mathbf{x} < 0$ for any $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$;
- ③ ... **positive semi-definite**, iff $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ for any $\mathbf{x} \in \mathbb{R}^n$;
- ④ ... **negative semi-definite**, iff $\mathbf{x}^T \mathbf{A} \mathbf{x} \leq 0$ for any $\mathbf{x} \in \mathbb{R}^n$;
- ⑤ ... **indefinite**, iff $\exists \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ s.t. $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$ and $\mathbf{y}^T \mathbf{A} \mathbf{y} < 0$.

Quadratic forms

The next theorem gives a very simple way to find the sign of the quadratic form, and makes a very tight connection with diagonalization.

Theorem 8 (Spectral characterization of definiteness)

Let $\mathbf{A}$ be an $n \times n$ real symmetric matrix. The matrix $\mathbf{A}$ is ...

- ① ... positive definite iff all its eigenvalues are positive;
- ② ... negative definite iff all its eigenvalues are negative;
- ③ ... positive semi-definite iff all its eigenvalues are non-negative;
- ④ ... negative semi-definite iff all its eigenvalues are non-positive;
- ⑤ ... indefinite iff it has both positive and negative eigenvalues.

Quadratic Forms

- Proving the previous theorem is simple enough and we will do it.
- However, for handmade computations, computing all eigenvalues may not be simple without a computer, so the following result is useful.

Theorem 9 (Sylvester's Criterion)

Let $\mathbf{A}$ be a real symmetric $n \times n$ matrix. Then:

- ① $\mathbf{A}$ is positive definite iff $\det(\mathbf{A}_k) > 0$ for all $k = 1, 2, \dots, n$, where A_k is the $k \times k$ leading principal submatrix of A
- ② $\mathbf{A}$ is negative definite iff $\det(\mathbf{A}_k)(-1)^k > 0$ for all $k = 1, 2, \dots, n$.
- ③ $\mathbf{A}$ is positive semi-definite iff $\det(\mathbf{A}_k) \geq 0$ for all $k = 1, 2, \dots, n$.
- ④ $\mathbf{A}$ is negative semi-definite iff $\det(\mathbf{A}_k)(-1)^k \geq 0$ for all $k = 1, 2, \dots, n$.
- ⑤ $\mathbf{A}$ is indefinite if $\det(\mathbf{A}_k) > 0$ and $\det(\mathbf{A}_j) < 0$ for some k, j .

Dynamic systems**

- These concepts often appear in dynamic models in economics.
- For example, suppose your model is linear and can be written as $\mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1}$ where $\mathbf{x} \in \mathbb{R}$ and $\mathbf{A}$ is $n \times n$ matrix.
- Clearly $\mathbf{x}_t = \mathbf{A}^t \mathbf{x}_0$, but we want to analyze the behavior of $\mathbf{x}_t$ across large periods of time.
- As it turns out, it is much simpler to analyze by using the link to diagonalization, since $\mathbf{A}^t = (\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1})^t = \mathbf{P}\mathbf{\Lambda}^t\mathbf{P}^{-1}$.
- We can fully infer $\mathbf{x}_t$ behavior from the eigenvalues of $\mathbf{A}$. For example, if the modulus/absolute value of all eigenvalues is less than 1, then $\mathbf{\Lambda}^t \rightarrow \mathbf{0}_n$ (the zero $n \times n$ matrix) as $t \rightarrow \infty$, which implies $\mathbf{x}_t \rightarrow \mathbf{0}$.
- If your dynamic system is not linear, then you can use some linearization technique that we (may) learn to approximate the model to a (linear) first order and simplify the analysis to simple linear algebra. You will see many economics textbooks and papers do this all the time for this convenience.

More decompositions**

Theorem 10 (LDL decomposition)

Let A be an $n \times n$ real symmetric matrix. Then A is positive definite iff there exists a real diagonal matrix D with positive entries on its diagonal and a real lower triangle matrix L with all 1's on its diagonal, such that $A = LDL^T$.

If we define $P := L\sqrt{D}$, where $\sqrt{D}$ is the diagonal matrix whose entries on its diagonal are the square root of the corresponding entries of D , then we have $A = PP^T$. This is called the **Choleski decomposition** of A .

Theorem 11 (Choleski decomposition)

Let A be an $n \times n$ real symmetric matrix. Then A is positive definite iff there exists a real lower triangle matrix P with all positive entries on its diagonal such that $A = PP^T$.